Verbatim Training Data Extraction Attacks on Autoregressive Language Models (Carlini et al. 2021)

Black-box sampling and membership inference metrics allow adversaries to extract verbatim training sequences—including personally identifiable information—directly from pre-trained language model parameters.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Adversaries can extract verbatim text sequences—including names, email addresses, phone numbers, source code snippets, and copyrighted text—from autoregressive language models by generating high-volume samples and ranking them via differential perplexity heuristics.

EVIDENCE AND CAUSAL ASSESSMENT

Controlled extraction experiments comparing perplexity under the target model to perplexity under a smaller reference model or zlib compression ratios isolate memorization as a causal function of repetition in the training corpus and model parameter capacity.

EFFECT SIZE AND PERSISTENCE

Sequences repeated even a modest number of times (e.g., 10–50 occurrences) in multi-billion-token corpora are memorized near-verbatim. Larger models memorize a substantially higher fraction of their training sets than smaller models trained on the same data.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as robust (<20%). Extraction success is heterogeneous across token frequencies: highly repeated boilerplate or web headers are trivially extracted, whereas single-occurrence sequences are harder to retrieve without targeted prefix prompting.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

A model failing to output private data under standard prompting does not imply that the data was unlearned; adversarial suffix prompting and representation probing can frequently resurrect suppressed training spans.

WHAT ACTION IS REASONABLY JUSTIFIED

Aggressively deduplicate and scrub sensitive PII from pre-training web corpora prior to training, and implement differential privacy (DP) or verifiable unlearning protocols where privacy guarantees are legally mandated.

VERIFIED SOURCES & DOI IDENTIFIERS

- Carlini, N., Tramer, F., Wallace, E., Jagielski, M., Herbert-Voss, A., Lee, K., ... & Raffel, C. (2021). Extracting Training Data from Large Language Models. USENIX Security Symposium 2021. https://arxiv.org/abs/2012.07805